

Missouri University of Science and Technology

Intro to Data Science
CS 3402 (Spring 2026)

Lecture 11 Group Project Topics

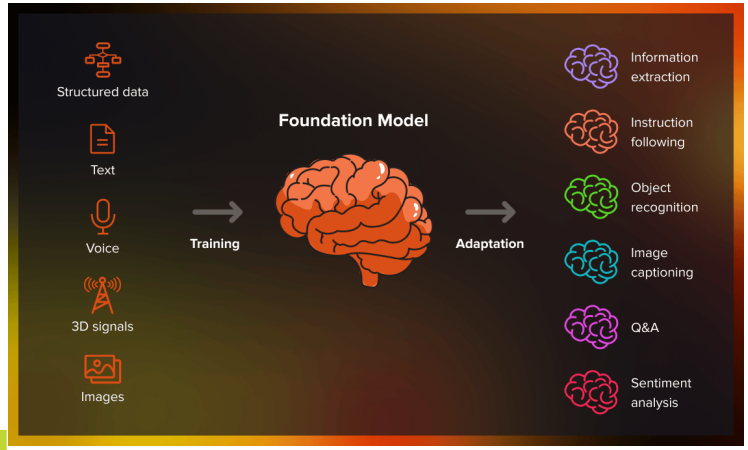
Xiaowei Yu, Ph.D.

Department of Computer Science

The Pretraining Paradigm

- Definition of foundation models

Foundation models (FMs) are **large-scale models** trained on **massive**, **diverse** datasets using (weakly-supervised) **self-supervised learning**, designed to **generalize** across multiple downstream tasks.



Source: serokell.io



Source: Visual Studio Magazine

The Pretraining Paradigm

- Rise of foundation models

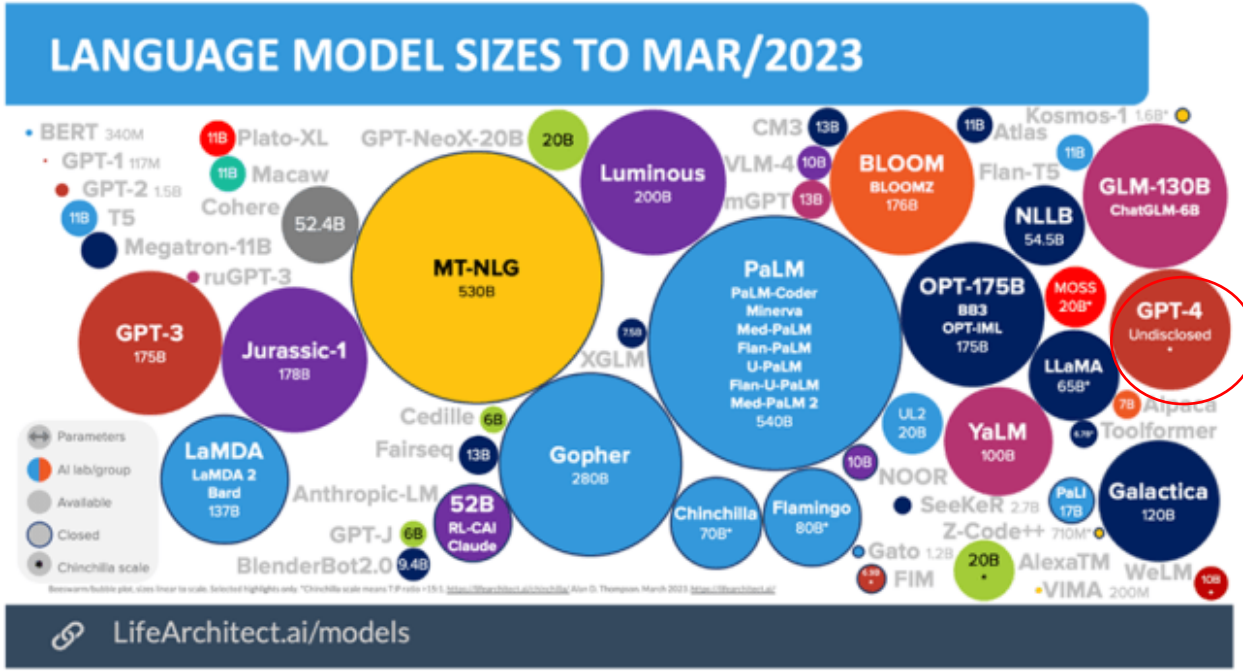
  From Task-Specific AI → General-Purpose AI

Type	Input	Key Tasks
Language	Text	QA, Translation, Summarization
Vision	Images	Classification, Generation, Segmentation
Multimodal	Text + Images/Video/Audio	Captioning, Cross-modal QA

The Pretraining Paradigm

- Rise of foundation models

Language Foundation Models

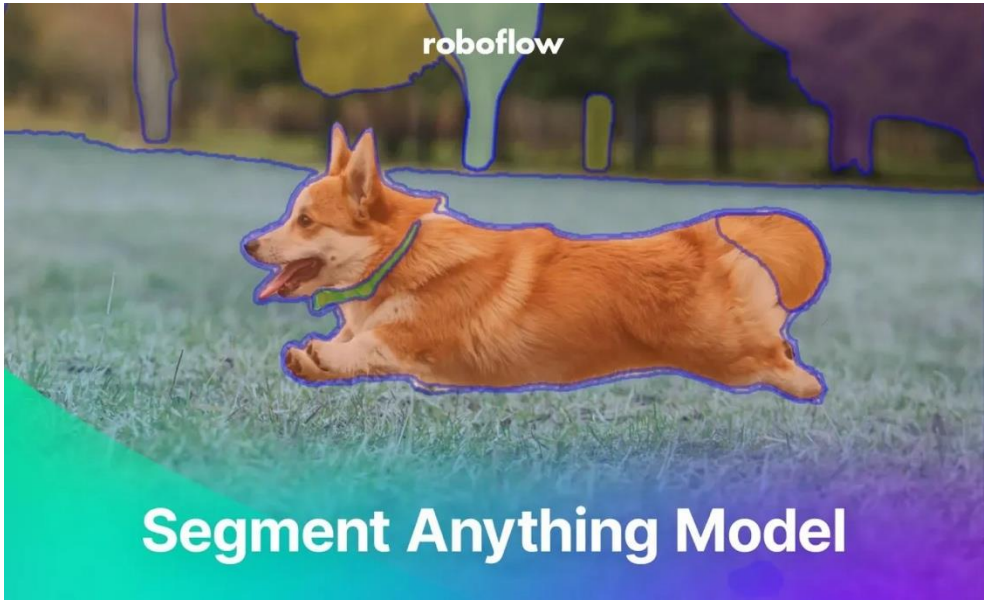


VLM

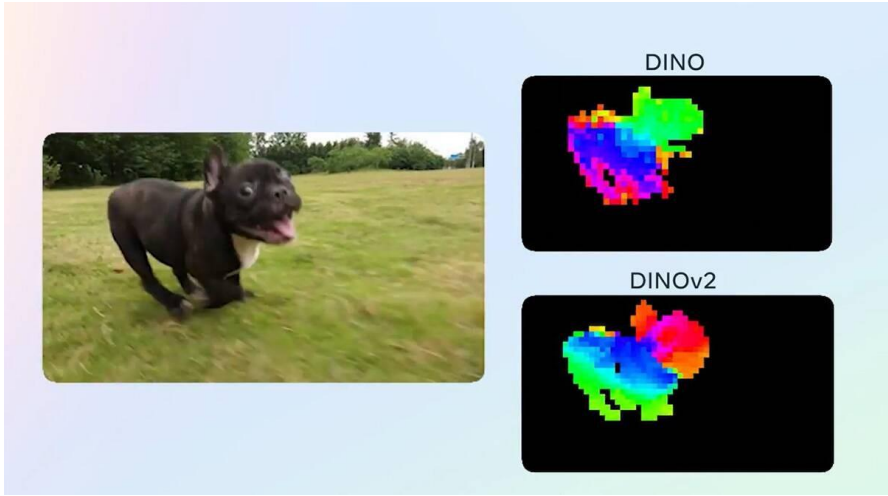
The Pretraining Paradigm

- Rise of foundation models

Vision Foundation Models



~450M for vision encoder; SA-1B dataset (~1 billion masks, 11 million images)



DINOv2 (Distillation with No Labels, version 2) ~1B images; ~21M to ~1.1B parameters

The Pretraining Paradigm

- Rise of foundation models

  Multimodal Foundation Models

ChatGPT

Gemini



Parameters: Unknown
Training dataset: Unknown

Inputs: Text / Images / Audios
Outputs: Text (answers, descriptions, explanations)
Combines **multimodal representation learning** in one model

The Pretraining Paradigm

- Success in NLP and Vision

Domain	Pretraining Data	Pretraining Task	Downstream Tasks	Example
NLP	Text corpora	Language modeling / Masked word prediction	Translation, QA, Summarization	GPT, BERT
Vision	Image datasets	Image classification / Self-supervised embedding	Segmentation, Detection, Retrieval	DINOv2, SAM,

- Learns **general patterns and representations** that can be reused.
- Reduces the need for **large labeled datasets** for every task.

Fine-tuning or prompting is usually required for **best performance** on specific tasks.

Group Project Topic 1

Question: Are general-purpose pretrained models better than task-specific models for medical datasets?

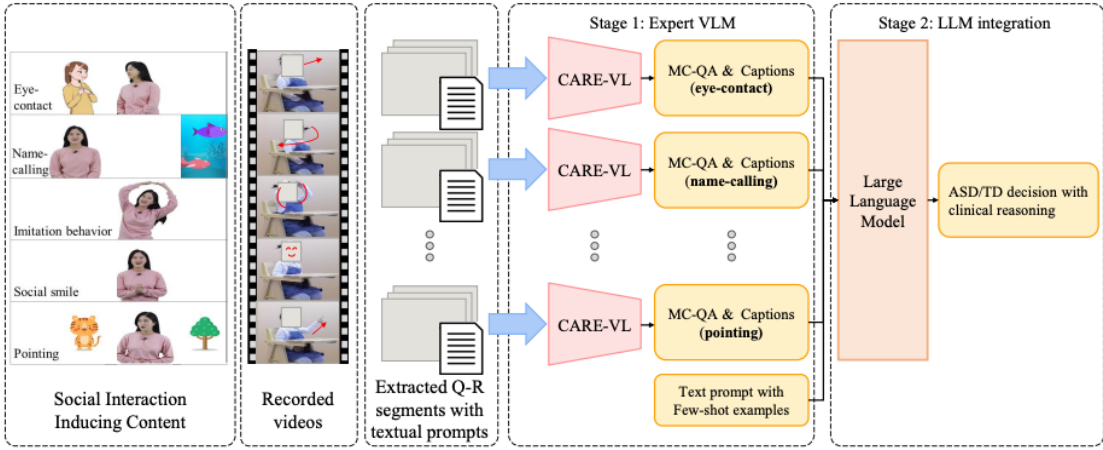
Intro: In recent years, large **pretrained foundation models**, such as LLMs, VLMs, have transformed machine learning by demonstrating strong generalization across a wide range of tasks, often outperforming models trained from scratch even when labeled data are limited. This success has motivated growing interest in applying general-purpose pretrained models to **medical and healthcare data**, where annotation is expensive and datasets are often small.

Group Project Topic 1

Question: Are general-purpose pretrained models better than task-specific models for medical datasets?



LLMs



CARE-VL: Vision-Language Model for ASD Screening

Group Project Topic 1

At the same time, many medical machine learning applications have traditionally relied on **task-specific model designs** that incorporate domain knowledge, such as structured features, biologically motivated representations, or specialized network architectures. These models embed strong **inductive biases** tailored to particular diseases, modalities, or data structures, which can be especially valuable in noisy and heterogeneous clinical datasets.

Whether pretrained foundation models consistently outperform task-specific models in medical applications remains an open and actively debated question. Medical datasets differ substantially from natural language or image benchmarks: they are typically smaller, highly structured, multimodal, and closely tied to underlying biological mechanisms. As a result, it is unclear when the benefits of large-scale pretraining outweigh the advantages of carefully designed, domain-specific modeling choices.

Group Project Topic 1

In this project, you will explore this question by **comparing task-specific designed models with pretrained or general-purpose foundation models across multiple medical datasets spanning different diseases and data modalities**. You will select medical datasets (**at least two**) and evaluate models tailored explicitly to the dataset alongside pretrained models adapted through transfer learning. The goal is to systematically assess their **predictive performance, robustness, and interpretability**, and to understand under what conditions domain-specific inductive bias may still outperform scale and pretraining.

Through these comparisons, you will gain hands-on experience with modern medical machine learning pipelines and develop critical insight into the tradeoffs between specialization and generalization in real-world healthcare applications.

Topic 1- Task-Specific Models vs. Pretrained Foundation Models for Medical Data Analysis

1. Data Collection and Preprocessing

You should select **two or more medical datasets**, potentially spanning different diseases or data modalities (e.g., medical images, clinical tabular data, signals, or text). Public sources such as **Kaggle**, **UCI Machine Learning Repository**, or other reputable open medical datasets may be used.

Dataset description: Disease; Data modality (image, tabular, time series, text, etc.); Source of the dataset
Data inspection and quality check: Identify missing values, incomplete records, or corrupted samples; Detect potential outliers or abnormal samples
Data preprocessing: Handle missing data (e.g., removal, imputation, masking); Normalize or standardize features if necessary; Resize, augment, or transform data for model compatibility (especially for images)
Dataset statistics: Total number of samples; Number of classes (if classification); Class distribution
Data splitting: Clearly describe how the data are split into training, validation, and test sets

All preprocessing choices should be **clearly documented and justified**.

Topic 1- Task-Specific Models vs. Pretrained Foundation Models for Medical Data Analysis

2. Model Selection and Experimental Design

You must compare **pretrained foundation models** with **task-specific designed models** on the same dataset(s).

(a) Large Language Models (LLMs)

For datasets involving text/structured data that can be converted into text form:
Examples: **ChatGPT, Claude, Gemini**, or similar models
Learning settings:
Zero-shot learning: no task-specific examples provided
Few-shot learning: a small number of labeled examples provided as prompts
Clearly describe:
Prompt design
Number of examples used
Evaluation method

Topic 1- Task-Specific Models vs. Pretrained Foundation Models for Medical Data Analysis

2. Model Selection and Experimental Design

You must compare **pretrained foundation models** with **task-specific designed models** on the same dataset(s).

(b) Pretrained Vision Models (if applicable)
For image-based medical datasets:
Examples: **ResNet**, **Vision Transformer (ViT)**, or other pretrained vision models
Learning settings:

- **Zero-shot or frozen-feature evaluation**
- **Fine-tuning**

Report:

- Training hyperparameters
- Performance Metrics

Topic 1- Task-Specific Models vs. Pretrained Foundation Models for Medical Data Analysis

2. Model Selection and Experimental Design

You must compare **pretrained foundation models** with **task-specific designed models** on the same dataset(s).

(c) Task-Specific Designed Models

You should design and train **task-specific models from scratch**, tailored to the chosen dataset.
Possible model choices include (but are not limited to):
MLP for tabular or low-dimensional data
CNN-based architectures for image or signal data
Transformer-based architectures trained from scratch
Other reasonable architectures motivated by the dataset structure
You should:
Explain the motivation behind the model design
Train models using only the task-specific data
Apply appropriate loss function and validation strategies

Topic 1- Task-Specific Models vs. Pretrained Foundation Models for Medical Data Analysis

3. Evaluation and Analysis

You should evaluate models using appropriate performance metrics, such as:
Accuracy, precision, recall, F1-score
ROC-AUC or PR-AUC (if applicable)
Other task-relevant metrics
Beyond raw performance, students are encouraged to analyze:
Robustness (e.g., sensitivity to hyperparameters, data size or noise)
Interpretability (feature importance, attention maps, or qualitative analysis)

Topic 1- Task-Specific Models vs. Pretrained Foundation Models for Medical Data Analysis

4. Documentation, Reporting, and Presentation

4.1 Project Report

You must submit a written report that includes:

- Introduction and motivation
- Dataset description and preprocessing
- Model descriptions and experimental setup
- Results and comparisons
- Discussion and key findings
- Limitations and future directions
- All results should be clearly organized using tables and figures.

Topic 1- Task-Specific Models vs. Pretrained Foundation Models for Medical Data Analysis

4. Documentation, Reporting, and Presentation

4.2 Presentation Slides

You should prepare a slide presentation covering:

- Problem definition and dataset
- Model comparison strategy
- Key experimental results
- Main insights and conclusions
- Presentations should emphasize **what was learned**, not just which model performed best.

Peer Evaluation of Presentations

- In addition to instructor evaluation, each team's presentation will be **rated by other teams** based on its **overall quality, clarity, and organization**.
- Please ensure your slides are well-structured and easy to follow, Clear visuals and a logical flow are strongly encouraged.

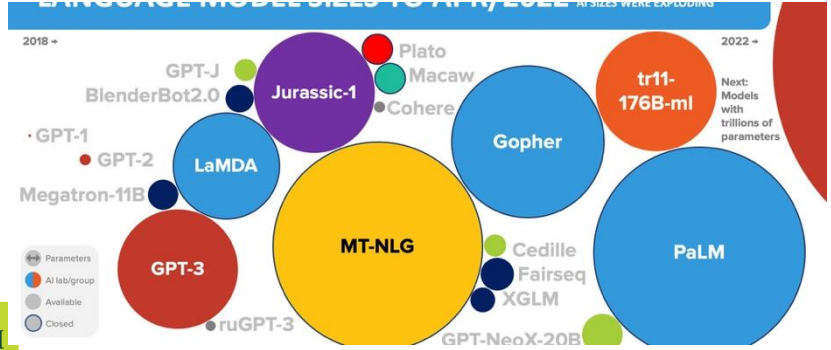
Group Project Topic 2 - LLMs

Title: Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

Background: Large Language Models (LLMs) such as ChatGPT and Gemini, are trained on massive text corpora collected from the internet. However, these training datasets are heavily dominated by English and high-resource regions, potentially leading to uneven performance across languages and countries. As a result, LLMs may produce responses that vary in accuracy, completeness, reasoning quality, or depth depending on the language or regional context of the prompt.

Group Project Topic 2

Aim: This project aims to systematically evaluate whether LLM performance is consistent across different languages (and potentially across countries or regions). By designing controlled prompts and comparing model responses using quantitative metrics such as accuracy, length, sentiment, and reasoning quality, students will analyze whether measurable discrepancies exist. This project highlights important issues in data representation, model generalization, and fairness in AI systems.





Group Project Topic 2

In this project, you will explore whether **large language models (LLMs)** perform consistently across different languages and regional contexts. You will design controlled prompts (e.g., factual questions, reasoning tasks, or knowledge-based queries) and evaluate model responses in multiple languages and/or across references to different countries. At least two languages (or regions) should be included to enable systematic comparison. The goal is to quantitatively assess differences in response accuracy, completeness, reasoning quality, sentiment, or other measurable characteristics, and to determine whether observable performance discrepancies exist.

Through this investigation, you will gain hands-on experience in experimental design for AI evaluation, prompt construction, quantitative metric development, and statistical comparison. More broadly, you will develop critical insight into how data distribution, representation imbalance, and training corpora composition may influence model behavior, robustness, and fairness in global AI systems.

Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

1. Data Collection and Preprocessing

You should design a structured prompt dataset to evaluate LLM performance across **at least two** languages (and optionally across different countries or regions). Prompts may include factual knowledge questions, reasoning tasks (e.g., math or logic), short explanatory questions, or domain-specific queries. Public question sources such as multilingual QA datasets, Wikipedia-based fact queries, etc.

Dataset description: Task type (factual QA, reasoning, explanation, sentiment, etc.); Languages included (e.g., English, Spanish, Chinese, etc.); Optional regional/country references (if applicable); Source of prompts (public dataset, translated version, or self-constructed).
Data inspection and quality check: Ensure semantic equivalence across languages (careful translation or back-translation if necessary); Remove ambiguous or culturally dependent questions that may introduce confounding factors; Verify correctness of reference answers (for objective tasks);
Data preprocessing: Ensure consistent instructions across languages; If translation is used, clearly document the translation method (human vs. automated).
Dataset statistics: Number of task categories; Number of prompts per language;

Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

2. Model Selection and Experimental Design

You must evaluate multiple large language models (LLMs) across multiple languages (and optionally across regional or country-based prompts) under controlled experimental settings.

(a) Pretrained Large Language Models (LLMs) - Closed-source models

You should select widely used pretrained foundation model of closed-source. Comparing multiple models is encouraged.

Examples: **ChatGPT, Claude, Gemini,** or similar models

Learning settings. You should evaluate model performance under consistent and clearly defined settings:

Zero-shot learning: no task-specific examples provided

Few-shot learning: a small number of labeled examples provided as prompts

To ensure a fair comparison across languages or regions, you must clearly describe:

Prompt design: Exact prompt templates used. Method of translation (manual, professional, automated).

Number of prompts: Total prompts per language. Number of prompts per task type.

Evaluation method: Whether human evaluation is used (if applicable). Whether responses are evaluated automatically (e.g., exact match, BLEU, accuracy).

Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

2. Model Selection and Experimental Design

You must evaluate multiple large language models (LLMs) across multiple languages (and optionally across regional or country-based prompts) under controlled experimental settings.

(b) Pretrained Large Language Models (LLMs) - Open-source models

You should select widely used pretrained foundation model of open source models. Comparing multiple models is encouraged.

Examples: **Open-source LLMs** (e.g., LLaMA-based models, Falcon-40B, Mistral) or similar models

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Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

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(c) Fine-Tuning Small Large Language Models

In addition to pretrained models, you are required to fine-tune at least one smaller open-source LLM (e.g., **GPT-2**, **LLaMA-1B**, or similar) on a subset of the prompt dataset.

- Collect at least one text dataset appropriate for your task (factual QA, reasoning, or explanatory prompts).
- Adapt the model to the selected task using the training subset.
- Reserve a separate set of prompts not used in fine-tuning to enable fair performance comparison.
- Clearly report all hyperparameters and settings, including learning rate, batch size, number of epochs, optimizer, and any data preprocessing applied.

Goal: Compare the performance of the fine-tuned small LLM with zero- or few-shot pretrained LLMs to analyze how task-specific adaptation affects accuracy, reasoning, and cross-linguistic consistency.

Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

3. Evaluation and Analysis

For all models (pretrained and fine-tuned):

Prompt design: clearly specify templates, translations, and examples used.
Number of prompts: total prompts per language, per task.
Metrics: accuracy, BLEU/ROUGE, response length, reasoning quality, sentiment, etc.
Statistical comparison: paired tests to quantify cross-linguistic performance differences.

Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

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Topic 2 - Evaluating Cross-Linguistic Performance Discrepancies in Large Language Models

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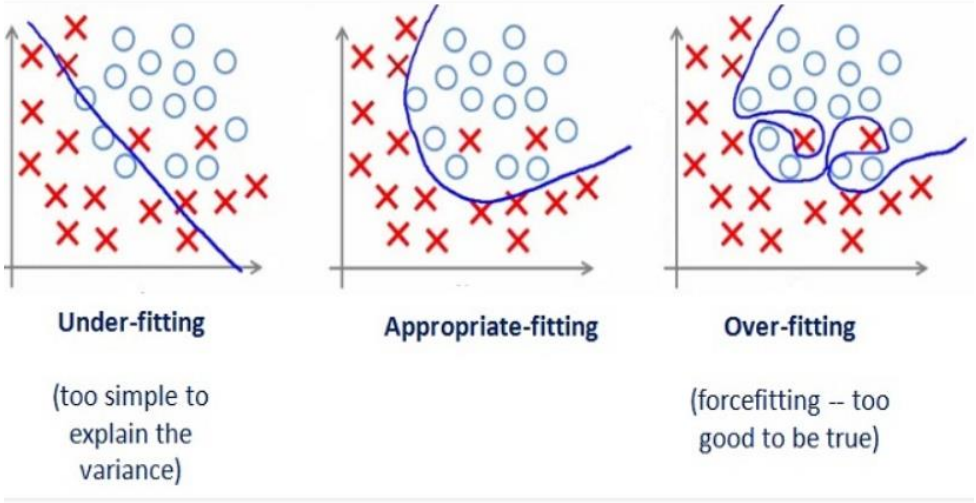
Group Project Topic 3

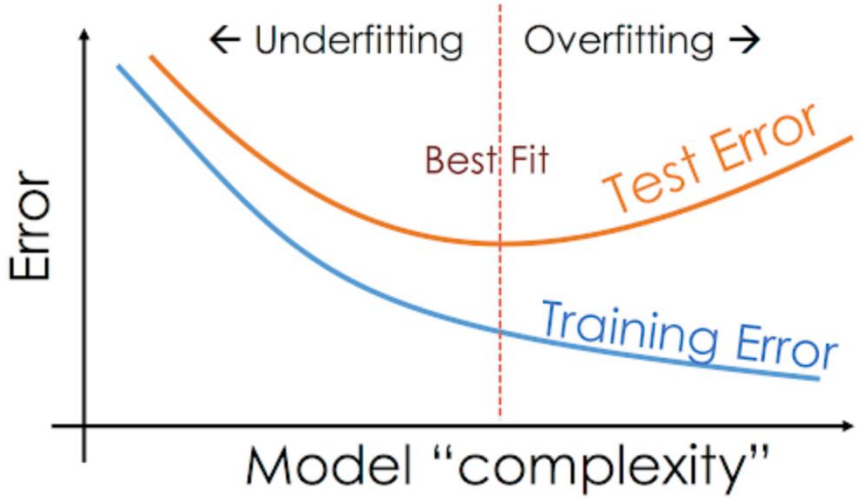
Title: Investigating the Impact of Training Data Size on Model Performance and Generalization

Background: In machine learning, model performance strongly depends on the amount of available training data. While larger datasets often improve generalization, the relationship is not always linear. Limited data may lead to high variance (overfitting), while overly complex models trained on small datasets may generalize poorly. Conversely, simple models may underfit even with more data.

Understanding how data size affects model stability, bias–variance tradeoff, and generalization gap is fundamental in data science.

Overfitting / Underfitting





In practice, finding the absolute "global optimum" is like chasing a ghost—you can never be 100% sure you're there. Instead, machine learning is really the art of **error management.**

Overfitting / Underfitting

Model State	Bias	Variance	Complexity	Description
Underfitting	**High**	**Low**	Too Low	The model "misses" the point (High Bias) but is consistent in its wrongness (Low Variance). Model is too rigid to learn the pattern.
Overfitting	**Low**	**High**	Too High	The model "hits" every training point (Low Bias) but changes wildly with new data (High Variance). Model memorizes noise and fails on new data.
Optimal Fit	**Low**	**Low**	Balanced	The "Sweet Spot" where Total Error is minimized.

Group Project Topic 3 - Investigating the Impact of Training Data Size on Model Performance and Generalization

This project aims to systematically evaluate how different training data sizes influence:

- Model accuracy and stability
- Overfitting and underfitting behavior
- Generalization gap (training vs. testing performance)

You will train models using different proportions of the dataset (10%, 30%, 50%, 100%) and analyze performance trends using quantitative metrics and learning curves.

Group Project Topic 3 - Investigating the Impact of Training Data Size on Model Performance and Generalization

1. Data Collection and Preprocessing

You must select a supervised learning dataset (classification or regression).
Possible Sources: Kaggle; UCI Machine Learning Repository; OpenML

Dataset Requirements: At least 3 different datasets; At least 1,000 samples per dataset; Clearly defined input features and labels;

Data inspection and quality check: Report dataset size (samples × features); Check for missing values; Detect outliers (if applicable); Perform necessary preprocessing, such as feature scaling (normalization/standardization), Train/test split
Data preprocessing: Ensure consistent instructions across languages; If translation is used, clearly document the translation method (human vs. automated).
Data splits for experiment: Keep the test set fixed. Create training subsets using: 10% of training data; 30%, 50%, 100%

Group Project Topic 3 - Investigating the Impact of Training Data Size on Model Performance and Generalization

2. Model Selection and Experimental Design

You must compare **at least two models.**

- One classical machine learning (ML) model (lower complexity)
- One Artificial Neural Network (ANN)-based model (higher complexity)

(a) Classical Machine Learning Model, such as Logistic Regression, Linear Regression, Support Vector Machine;

(b) Artificial Neural Network (ANN)-based model, such as Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN), Graph Neural Network (GNN), Transformer.

For both models: Use the same fixed test set; Train with: 10% data, 30% data, 50% data, 100% data. Repeat each experiment at least 3 times; Report mean ± standard deviation.

Group Project Topic 3 - Investigating the Impact of Training Data Size on Model Performance and Generalization

3. Evaluation and Analysis

3.1 Metrics

Classification:
Accuracy
Precision / Recall / F1-score
Confusion matrix

Regression:
MSE / RMSE
R^2 score

For all models:
Training error
Testing error

Group Project Topic 3 - Investigating the Impact of Training Data Size on Model Performance and Generalization

3. Evaluation and Analysis

3.2 Learning Curves

Plot:
Training performance vs. data size
Testing performance vs. data size

Analyze:

- Does performance saturate?
- Does the generalization gap shrink?
- Is the model data-hungry?

Group Project Topic 3 - Investigating the Impact of Training Data Size on Model Performance and Generalization

3. Evaluation and Analysis

3.3 Interpret results

Does additional data reduce overfitting?
Which model benefits more from increased data?

Small data + complex model → High variance?
Large data + simple model → High bias?

Group Project Topic 3 - Investigating the Impact of Training Data Size on Model Performance and Generalization

4. Documentation, Reporting, and Presentation

4.1 Project Report

- Introduction and motivation
- Dataset description and preprocessing steps
- Model descriptions
- Experimental setup
- Learning curve plots
- Tables comparing performance
- Bias–variance discussion
- Key conclusions
- Limitations and future directions

Group Project Topic 3 - Investigating the Impact of Training Data Size on Model Performance and Generalization

4. Documentation, Reporting, and Presentation

4.2 Presentation Slides

- Research question
- Dataset and preprocessing
- Model comparison
- Learning curves
- Generalization gap analysis
- Bias–variance insights
- Final conclusions

Peer Evaluation of Presentations

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Group Project Topic 4 - LLMs

Title: Evaluating Prompt Engineering vs Fine-Tuning for Domain Adaptation in Large Language Models

Background: Large Language Models (LLMs) such as ChatGPT, LLaMA, and Mistral are pretrained on massive text corpora covering general knowledge. However, their ability to adapt to specific domains (e.g., medicine, law, finance, or STEM subjects) may vary depending on the adaptation strategy. Two common approaches are:

- **Prompt Engineering:** Designing specific prompts (zero-shot or few-shot) without changing model weights.
- **Fine-Tuning:** Updating model weights on a task-specific dataset to adapt to the domain.

Which strategy is more effective for LLMs?

Group Project Topic 4 - Evaluating Prompt Engineering vs Fine-Tuning for Domain Adaptation in Large Language Models

Systematically evaluate whether prompt engineering or fine-tuning is more effective for domain adaptation in LLMs. You will:

- Compare zero-shot, few-shot, and fine-tuned adaptation strategies
- Quantitatively measure performance on factual, reasoning, or explanatory tasks
- Analyze trade-offs in accuracy, reasoning quality, cost, and compute requirements

Group Project Topic 4 - Evaluating Prompt Engineering vs Fine-Tuning for Domain Adaptation in Large Language Models

1. Data Collection and Preprocessing

Dataset Requirements: At least 3 different datasets;
Task types: factual QA, reasoning, explanation, sentiment, etc.
Source of prompts: public datasets (e.g., MedQA, FinQA), translated or self-constructed prompts

Data inspection and quality check: Remove ambiguous or overly general prompts; Preprocess to standardize format and instruction phrasing
Dataset statistics: Number of tasks; Number of prompts per task type

Group Project Topic 4 - Evaluating Prompt Engineering vs Fine-Tuning for Domain Adaptation in Large Language Models

2. Model Selection and Experimental Design
You must evaluate multiple large language models (LLMs) under controlled experimental settings.

(a) Large Language Models (LLMs) - Closed-source models
You should select widely used pretrained foundation model of closed-source. Comparing multiple models is encouraged.
Examples: **ChatGPT, Claude, Gemini,** or similar models
Learning settings. You should evaluate model performance under consistent and clearly defined settings:
Zero-shot learning: no task-specific examples provided
Few-shot learning: a small number of labeled examples provided as prompts

Prompt design: Exact prompt templates used.

Group Project Topic 4 - Evaluating Prompt Engineering vs Fine-Tuning for Domain Adaptation in Large Language Models

2. Model Selection and Experimental Design
You must evaluate multiple large language models (LLMs) under controlled experimental settings.

(b) Large Language Models (LLMs) - Open-source models
You should select widely used pretrained foundation model of open source models. Comparing multiple models is encouraged.
Examples: **Open-source LLMs** (e.g., LLaMA-based models, Falcon-40B, Mistral) or similar models
Learning settings. You should evaluate model performance under consistent and clearly defined settings:
Zero-shot learning: no task-specific examples provided
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Group Project Topic 4 - Evaluating Prompt Engineering vs Fine-Tuning for Domain Adaptation in Large Language Models

2. Model Selection and Experimental Design
You must evaluate multiple large language models (LLMs) under controlled experimental settings.

(c) Fine-Tuning Small LLMs
Fine-tune a smaller model (e.g., OpenAI GPT-2, Google T5, Meta LLamA-1B) on a **subset of domain-specific prompts**

Report all hyperparameters: learning rate, batch size, epochs, optimizer, hardware

Group Project Topic 4 - Evaluating Prompt Engineering vs Fine-Tuning for Domain Adaptation in Large Language Models

3. Evaluation and Analysis

For all models:

- **Accuracy / Exact Match** – factual correctness
- **Reasoning Quality** – logical consistency, step-wise correctness
- **Response Characteristics** – length, completeness, hallucination rate
- **Optional Metrics** – BLEU/ROUGE for generation tasks, sentiment, or style

Analysis:

- Compare zero-shot, few-shot, and fine-tuned performance
- Assess cost vs performance: compute time, training requirements

Group Project Topic 4 - Evaluating Prompt Engineering vs Fine-Tuning for Domain Adaptation in Large Language Models

4. Documentation, Reporting, and Presentation

4.1 Project Report

- Introduction and motivation
- Dataset description and preprocessing steps
- Model descriptions
- Experimental setup
- Learning curve plots
- Tables comparing performance
- Key conclusions
- Limitations and future directions

Group Project Topic 4 - Evaluating Prompt Engineering vs Fine-Tuning for Domain Adaptation in Large Language Models

4. Documentation, Reporting, and Presentation

4.2 Presentation Slides

- Research question
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Group Project Topic 5 – Multimodal Models

Multimodal Learning vs Single-Modality Models

Background: Many real-world problems involve **multiple types of data (modalities)**:

- Medical diagnosis → medical images + clinical variables
- Product prediction → text reviews + structured ratings
- Healthcare prediction → notes + lab results
- Vision tasks → images + captions

However, an important research question remains:
Does combining multiple modalities actually improve predictive performance compared to using a single modality?

Multimodal learning increases model complexity and computational cost. Therefore, it is important to empirically evaluate whether multimodal models truly provide measurable benefits.

Group Project Topic 5 – Multimodal Learning vs Single-Modality Models

Requirements:
Select datasets containing **at least two modalities**
Train:
- Model A: Modality 1 only
- Model B: Modality 2 only
- Model C: Combined modalities

Quantitatively compare performance

Analyze whether multimodal learning improves results

Perform statistical testing to verify differences

Group Project Topic 5 – Multimodal Learning vs Single-Modality Models

1. Data Collection and Preprocessing

At least 2 different datasets
Possible Dataset Types: Image + Text; Image + Tabular; Image + Phenotype; Image + Genotype

Dataset Description: Modalities included; Number of samples; Task definition; Data split

Data Cleaning: Standardize numerical features; normalization; Handle missing values

Group Project Topic 5 – Multimodal Learning vs Single-Modality Models

2. Model Selection and Experimental Design

You need to build **three models**:

Suppose you choose Image + Text

Model A — Image Only
- CNN (image only)
- Vision Transformer (image only)

Model B — Text Only
- Language Model (text only)

Model C — Image + Text
Multimodal Model, e.g., CLIP (Image + Text)

Group Project Topic 5 – Multimodal Learning vs Single-Modality Models

3. Evaluation and Analysis

For classification:

- Accuracy
- Precision
- Recall
- F1-score

Required Analysis Questions

Does multimodal learning improve performance?
How large is the improvement?
Does one modality dominate the prediction?
What are the computational costs of multimodal models? (GPU, Training Time, etc.)

Group Project Topic 5 – Multimodal Learning vs Single-Modality Models

4. Documentation, Reporting, and Presentation

4.1 Project Report

- Introduction and motivation
- Dataset description and preprocessing steps
- Model descriptions
- Experimental setup
- Learning curve plots
- Tables comparing performance
- Key conclusions
- Limitations and future directions

Group Project Topic 5 – Multimodal Learning vs Single-Modality Models

4. Documentation, Reporting, and Presentation

4.2 Presentation Slides

- Research question
- Dataset and preprocessing
- Model comparison
- Learning curves
- Performance Comparision
- Final conclusions

Peer Evaluation of Presentations

- In addition to instructor evaluation, each team's presentation will be **rated by other teams** based on its **overall quality, clarity, and organization**.
- Please ensure your slides are well-structured and easy to follow, Clear visuals and a logical flow are strongly encouraged.

You can propose your interested project topic. I can help modify your proposal to ensure it has an equivalent workload to these topics.